

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1 | #include <stdio.h>
2 | #include <stdlib.h>
3 |
4 | int find_pair_sum(int arr[], int low, int high, int x, int *el1, int *el2) {
5 |
6 |     if (low >= high) {
7 |         return 0;
8 |     }
9 |
10 |     int mid = low + (high - low) / 2;
11 |
12 |
13 |     if (find_pair_sum(arr, low, mid, x, el1, el2)) {
14 |         return 1;
15 |     }
16 |
17 |
18 |     if (find_pair_sum(arr, mid + 1, high, x, el1, el2)) {
19 |         return 1;
20 |     }
21 |
22 |     int i = low;
23 |     int j = high;
24 |     while (i <= mid && j > mid) {
25 |         int current_sum = arr[i] + arr[j];
26 |         if (current_sum == x) {
27 |             *el1 = arr[i];
28 |             *el2 = arr[j];
29 |             return 1;
30 |         } else if (current_sum < x) {
31 |             i++;
32 |         } else {
33 |             j--;
34 |         }
35 |     }
36 |
37 |     return 0;
38 | }
39 |
40 | int main() {
41 |     int n;
42 |     if (scanf("%d", &n) != 1 || n < 2) {
43 |         printf("No\n");
44 |         return 0;
45 |     }
46 |
47 |     int *arr = (int *)malloc(n * sizeof(int));
48 |     for (int i = 0; i < n; i++) {
49 |         if (scanf("%d", &arr[i]) != 1) {
50 |             free(arr);
51 |             return 0;
52 |         }

```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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